

Assignment:-2

Task 1:- List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply

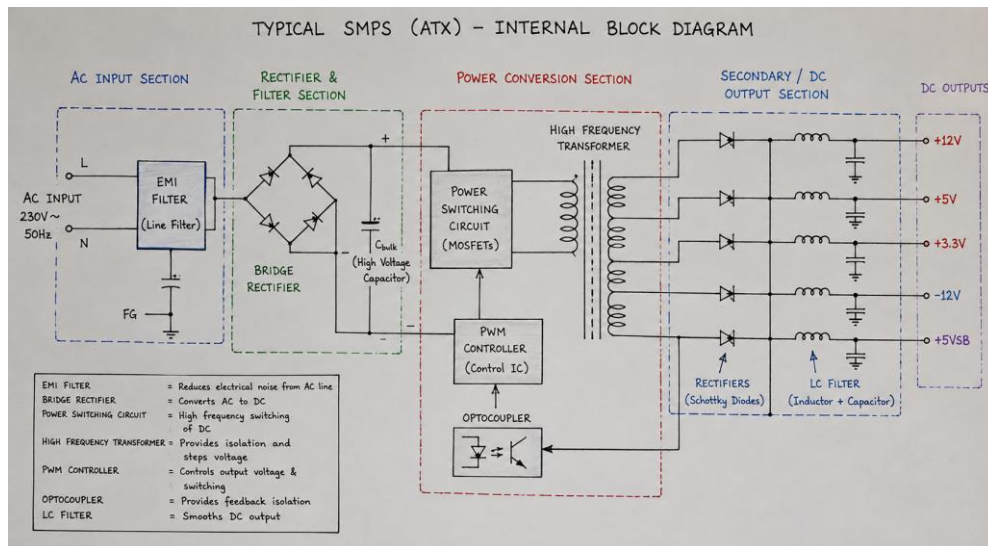
SMPS Devices

1. **Laptop charger** — Changes AC power into DC power for the laptop.
2. **LED TV** — Gives the TV the correct power safely and efficiently.
3. **Computer** — Provides different voltages needed by computer parts.

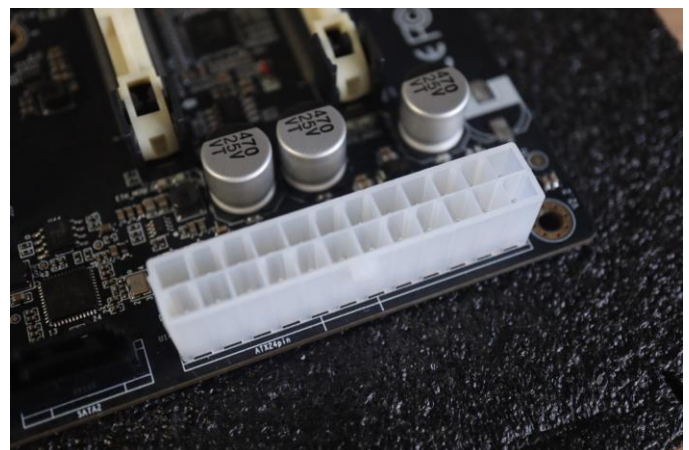
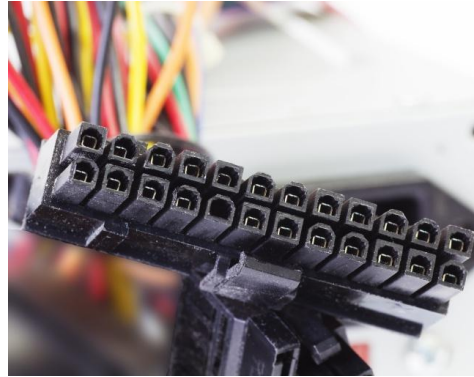
UPS Devices

1. **Computer** — Gives backup power when electricity goes off.
2. **Wi-Fi router** — Keeps the internet working during power cuts.

Task 2:- Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections. Hint: You can hand-draw and upload a photo, or use any free diagram tool like draw.io



Task 3:-Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.



Connector	Devices it powers	Safety tip
ATX 24-pin connector	Motherboard	Turn off and unplug the PC before connecting or removing it.
SATA power connector	HDD, SSD, DVD/optical drives	Match the L-shaped connector correctly; never force it.
Molex connector	Case fans and older drives/peripherals	Hold the connector itself when removing it; don't pull the wires.
PCIe 6/8-pin connector	Graphics card (GPU)	Make sure the clip locks into place and use the correct PCIe cable.

Task 4:- Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

Steps

1. Turn off the PC and unplug it from the wall.
2. Motherboard: Connect the 24-pin ATX cable to the motherboard.
3. CPU: Connect the 4/8-pin CPU cable near the processor.
4. Graphics card: Connect the 6/8-pin PCIe cable to the graphics card.
5. SSD/HDD: Connect the SATA power cable to the storage drive.
6. Check all cables and make sure they are firmly connected.
7. Plug in the PC and turn it on.